THE BOX-IN-SPHERE FRACTAL ATLAS (II)

Interaction-first mapping: phenomena → math branches

(A learning navigation tool; not a replacement for definitions and proofs)

Kevin T.N.

jkdkr2439@gmail.com

March 2026

Abstract

This is the **interaction-first** edition of the Box-in-Sphere Atlas. Instead of starting from branch names and then attaching metaphors, we start from **simple visible interactions** between a Box and a Sphere (sliding, rotating, squeezing, tilting, slicing, projecting, gluing, cutting, random shaking, multi-scale zoom). Only *after* the phenomenon is clear do we name the mathematical branch that formalizes it. The goal is to make the Box–Sphere imagery actually useful as a learning interface: you see an interaction, you recognize a math language.

Scope Oath (Contract). This atlas is **not** mathematics. It is a **navigation layer**. It helps you: (1) orient, (2) choose tools, (3) translate languages, (4) detect limits, (5) plan study routes. If a claim needs truth, return to formal work: definitions, axioms, proofs, models, computations.

Contents

1 Part 0 — How to Use This Atlas (Interaction-first rule) 3

2 Part I — The Stage: Box, Sphere, Regions, and Motion Types 3

2.1 The stage (Box inside Sphere) . . . 3

2.2 Four regions (where your mind is operating) . . . 3

2.3 Motion dictionary (the interactions you are allowed to perform) . . . 4

2.4 Two crucial camera choices (why students get confused) . . . 4

3 Part II — The Branch Modules (Phenomenon first, branch second) 4

3.1 (1) Proof Theory (Syntactic Logic) . . . 5

3.2 (2) Model Theory (Semantic Logic) . . . 5

3.3 (3) Set Theory / Foundations . . . 6

3.4 (4) Combinatorics / Graph Theory . . . 6

3.5 (5) Linear Algebra . . . 7

3.6 (6) Abstract Algebra (Groups / Rings / Fields) . . . 7

3.7 (7) Euclidean / Affine Geometry . . . 8

3.8 (8) Projective Geometry . . . 8

3.9 (9) Topology . . . 9
3.10 (10) Differential Topology . . . 9
3.11 (11) Differential Geometry (Riemannian) . . . 10
3.12 (12) Real / Complex Analysis . . . 10
3.13 (13) Functional Analysis . . . 11
3.14 (14) Measure Theory / Integration . . . 11
3.15 (15) Probability Theory . . . 12
3.16 (16) Stochastic Processes (SDE intuition) . . . 12
3.17 (17) ODE / Dynamical Systems . . . 13
3.18 (18) PDE / Applied Analysis . . . 13
3.19 (19) Numerical Analysis . . . 14
3.20 (20) Optimization / Variational Methods . . . 14
3.21 (21) Number Theory . . . 15
3.22 (22) Algebraic Geometry . . . 15
3.23 (23) Category Theory (Meta-language) . . . 16
3.24 (24) Computability Theory . . . 16
3.25 (25) Complexity Theory . . . 17

4 Part III — Fractal Extension (Multi-scale zoom that preserves the same logic) 17

5 Part IV — A Short "How to Study" Protocol (using interaction-first mapping) 18

6 Conclusion 18

1 Part 0 — How to Use This Atlas (Interaction-first rule)

Principle 1 (Interaction-first mapping rule)**.** For each branch, we proceed in this order:

1. Describe a **Box–Sphere interaction** in plain language.
2. Identify what stays the same (the **invariants**) and what is allowed to change (the **moves**).
3. Only then name the **math branch** that studies exactly those moves and invariants.

If the branch name comes first, the atlas becomes decorative. If the phenomenon comes first, the atlas becomes a tool.

Rule of thumb 1 (One sentence test)**.** If you cannot describe the topic as "what happens when I do *this* to the box inside the sphere", you are not yet grounded. Pause and build the scene first.

2 Part I — The Stage: Box, Sphere, Regions, and Motion Types

2.1 The stage (Box inside Sphere)

We imagine a Box inside a Sphere. You can move the Box, deform the Box, move the Sphere with it (or not), and watch what changes.

- **Box** = the part of knowledge that behaves like a strict contract: rule-following, finite checking, discrete structure.
- **Sphere** = the part that behaves like a smooth world: continuous variation, models, approximation, flows.

2.2 Four regions (where your mind is operating)

- **Inside the Box:** "I can justify each step locally."
- **Between Box and Sphere:** "I explore shapes, families, approximations."
- **Boundary:** "the current tool stops working or changes phase."
- **Outside:** "not representable under the current contract (yet or in principle)."

2.3 Motion dictionary (the interactions you are allowed to perform)

Core interactions you will reuse everywhere:

- **Slide (translation):** the box shifts position; shape stays the same.
- **Rotate:** the box turns; shape stays the same.
- **Zoom (uniform scale):** the box grows/shrinks evenly.
- **Squeeze/Stretch (anisotropic scale):** the box becomes longer in some directions.
- **Tilt (shear):** the top face drifts sideways; faces stay flat, but right angles break.
- **Project:** you cast the box onto a wall; you lose depth on purpose.
- **Slice:** you cut the world with a plane; you study the cross-section.
- **Twist:** different layers rotate by different amounts (top turns more than bottom).
- **Bend:** straight becomes curved without tearing.
- **Fold/Pinch:** you create creases; smoothness breaks at a seam.
- **Cut/Glue:** you change connectivity (holes, handles, split/merge).
- **Shake (random):** motion is not chosen by you; it is sampled.
- **Zoom-in fractal:** at a corner you find a smaller box-in-sphere again.

2.4 Two crucial camera choices (why students get confused)

Camera Choice A: Move the box, keep the sphere fixed. You see the box change position inside a stable world (absolute frame).
Camera Choice B: Move both box and sphere together. From inside, nothing seems to change: this is a change of reference frame.

3 Part II — The Branch Modules (Phenomenon first, branch second)

Rule of thumb 2 (How to read each module)**.** Each module follows the same five lines: **Scene (what you see). Name (the math that formalizes it). Allowed moves (what counts as a "legal" interaction). Typical tasks (what you practice). Boundary signal (how the tool fails). Tool-switch (what you do next).**

3.1 (1) Proof Theory (Syntactic Logic)

(1) Proof Theory

Scene (what you see). You are inside the box with a rulebook. You do not "see" the whole sphere. You only perform local legal moves: replace one small pattern by another small pattern. You can replay and check every move step by step.
Name (the math that formalizes it). Proof theory is the study of rule-driven motion inside the box. A proof is a path made of certified micro-steps.
Allowed moves (what counts as a "legal" interaction). Legal moves are *rewrites* (inference rules). Nothing continuous is assumed.
Typical tasks (what you practice). Build a path from assumptions to conclusion; normalize; eliminate detours; show that certain moves never produce contradictions.
Boundary signal (how the tool fails). *Proof boundary:* some true statements cannot be reached in a given rulebook.
Tool-switch (what you do next). If stuck, either strengthen the rulebook (change axioms) or switch to semantic truth (model theory) and re-state the goal.

3.2 (2) Model Theory (Semantic Logic)

(2) Model Theory

Scene (what you see). Instead of moving symbols, you place many different boxes inside the same sphere: each box is a different "world" (a different interpretation). You ask: which statements stay true no matter which world-box you pick?
Name (the math that formalizes it). Model theory formalizes the "many worlds inside one sphere" viewpoint.
Allowed moves (what counts as a "legal" interaction). Legal moves are *changing interpretation* while keeping the language fixed: rename, reduce, expand, compare structures.
Typical tasks (what you practice). Show a sentence holds in all models of a theory; build countermodels; compare expressivity.
Boundary signal (how the tool fails). *Representation boundary:* the language is too weak (or too costly) to express the phenomenon you want to describe.
Tool-switch (what you do next). Enrich the language, change the structure class, or return to syntax and prove what can be proved.

3.3 (3) Set Theory / Foundations

(3) Set Theory / Foundations

Scene (what you see). You build the box itself. You decide what counts as a "piece" of space, what collections are allowed, and which construction steps are permitted. You are not moving inside a given box; you are specifying the contract of the box.
Name (the math that formalizes it). Set theory is box-construction: the foundational contract for building objects.
Allowed moves (what counts as a "legal" interaction). Legal moves are *construction steps* licensed by axioms: forming sets from sets, iterating hierarchies, extending the universe by controlled techniques.
Typical tasks (what you practice). Build objects, compare sizes of infinities, analyze what follows from which axiom.
Boundary signal (how the tool fails). *Proof boundary:* some questions are independent of standard axioms.
Tool-switch (what you do next). Add/modify axioms, or work conditionally: "if this axiom, then that result".

3.4 (4) Combinatorics / Graph Theory

(4) Combinatorics / Graph Theory

Scene (what you see). The box is made of nodes and connections. You rewire edges, remove edges, contract edges, and watch what connectivity patterns survive. The sphere is not needed; the world is discrete.
Name (the math that formalizes it). Combinatorics and graph theory formalize discrete interaction and counting.
Allowed moves (what counts as a "legal" interaction). Legal moves are discrete edits: add/remove edges, contract parts, relabel, traverse.
Typical tasks (what you practice). Count configurations; prove invariants; show a certain pattern must exist.
Boundary signal (how the tool fails). *Computational boundary (sometimes):* the space of configurations explodes and exact counting becomes infeasible.
Tool-switch (what you do next). Use recursion, generating functions, probabilistic method, or approximate counting.

3.5 (5) Linear Algebra

(5) Linear Algebra

Scene (what you see). Take a perfect box in the sphere. Now you deform it in the *straightest possible way*: faces stay flat, edges stay straight, but the box can be stretched, squeezed, tilted, or rotated. You are studying the "modes" of box deformation.
Name (the math that formalizes it). Linear algebra is the mathematics of straight box-deformation.
Allowed moves (what counts as a "legal" interaction). Legal moves are straight deformations: rotate, scale, squeeze by direction, shear, reflect, project (collapse depth), and change camera/basis (same deformation, different coordinates).
Typical tasks (what you practice). Solve intersection problems of many planes (where do they meet?); find stable directions that keep their orientation (axes that only grow/shrink); detect when the box collapses into a lower-dimensional sheet.
Boundary signal (how the tool fails). *Representation boundary:* the box becomes nearly flat or collapses, and naive coordinates become unstable or misleading.
Tool-switch (what you do next). Change basis (rotate your coordinate frame), factor the deformation into simpler steps, or project onto the relevant subspace (slice the motion).

3.6 (6) Abstract Algebra (Groups / Rings / Fields)

(6) Abstract Algebra

Scene (what you see). You stop caring about the physical box shape and care only about *rules of combination*. You pick two moves and define what it means to do one after the other. You ask: which rules must hold so that the world behaves consistently?
Name (the math that formalizes it). Abstract algebra is the study of rule-systems for combining moves.
Allowed moves (what counts as a "legal" interaction). Legal moves are *composition* and *structure-preserving translation*: turn one rule-world into another without breaking the rules.
Typical tasks (what you practice). Identify the "identity move" and the "undo move"; compare two rule-worlds by a structure-preserving map; compress a world by identifying moves that behave the same (quotient).
Boundary signal (how the tool fails). *Representation boundary:* the same underlying set can support many different rule-structures, so "same elements" does not mean "same algebra".
Tool-switch (what you do next). Track invariants of the rule-system, not the names of the elements.

3.7 (7) Euclidean / Affine Geometry

(7) Euclidean / Affine Geometry

Scene (what you see). You draw lines, planes, circles, and spheres in the world. You move the box around (slide/rotate) and ask what stays the same: distances, angles, parallel lines, alignments.
Name (the math that formalizes it). Geometry formalizes how shapes behave under chosen motion laws.
Allowed moves (what counts as a "legal" interaction). Legal moves depend on the chosen contract: rigid motions preserve distances; affine motions preserve straightness and parallelism.
Typical tasks (what you practice). Decide when two configurations are "the same" under the allowed motions; use slicing (cross-sections) to see hidden structure; place coordinates to compute intersections.
Boundary signal (how the tool fails). *Representation boundary:* coordinate choices hide invariants; you confuse the picture with the object.
Tool-switch (what you do next). Change viewpoint: pick invariants matched to the motion class (distance vs parallelism), or take clean slices (plane cuts) to simplify.

3.8 (8) Projective Geometry

(8) Projective Geometry

Scene (what you see). You look at the box through a camera. Parallel edges in the world appear to meet in the image. Some distinctions (distance, angle) are no longer stable under viewing. You upgrade the sphere to include "directions at infinity" so the camera-view becomes consistent.
Name (the math that formalizes it). Projective geometry is the math of camera-stable structure (incidence).
Allowed moves (what counts as a "legal" interaction). Legal moves are viewpoint transforms: what the camera can do. Straight lines remain straight, but measurements do not survive.
Typical tasks (what you practice). Explain why "parallel" is not absolute under perspective; use a completion so every pair of lines meets (including at infinity); track invariants that survive the camera.
Boundary signal (how the tool fails). *Representation boundary:* Euclidean intuitions break (angles/distances), but incidence remains.
Tool-switch (what you do next). Replace metric questions by incidence questions; treat "infinity" as a controlled boundary component.

3.9 (9) Topology

(9) Topology

Scene (what you see). You take the box and sphere and allow them to be made of soft rubber. You can bend and stretch freely, but you are not allowed to tear or glue. Now the only thing that matters is what is connected to what, and how many holes exist.
Name (the math that formalizes it). Topology is the mathematics of soft deformation.
Allowed moves (what counts as a "legal" interaction). Legal moves are bend/stretch without tearing or gluing.
Typical tasks (what you practice). Decide whether two shapes are the same under soft motion; count holes; detect whether a loop can shrink to a point.
Boundary signal (how the tool fails). *Type-change boundary:* the moment you cut or glue, you switch to a different topological type.
Tool-switch (what you do next). Use invariants (connected components, holes), not lengths and angles.

3.10 (10) Differential Topology

(10) Differential Topology

Scene (what you see). Same soft world as topology, but now you require the motion to be *smooth*: no sharp corners appear, unless you explicitly study where smoothness fails. You watch what happens when a smooth surface passes through a critical configuration.
Name (the math that formalizes it). Differential topology is topology with smoothness control.
Allowed moves (what counts as a "legal" interaction). Legal moves are smooth bend/stretch (smooth maps). "Bad" points become special events.
Typical tasks (what you practice). Understand critical points and how they change the global picture; use generic slicing to avoid accidental degeneracies; study how singular events reorganize the shape.
Boundary signal (how the tool fails). *Limit/regularity boundary:* smoothness breaks at singular maps or non-generic intersections.
Tool-switch (what you do next). Perturb slightly (genericity), slice transversely, and track invariants through controlled events.

3.11 (11) Differential Geometry (Riemannian)

(11) Differential Geometry (Riemannian)

Scene (what you see). You keep the sphere world smooth, but you now care about how the world measures itself: some directions "feel longer" than others. As you walk on the surface, the notion of straight line becomes "geodesic" (straight *according to the world*).
Name (the math that formalizes it). Differential geometry is the mathematics of measured smooth worlds.
Allowed moves (what counts as a "legal" interaction). Legal moves are smooth coordinate changes plus measurement-preserving descriptions. Bending is not only allowed; it becomes data (curvature).
Typical tasks (what you practice). Describe geodesics (natural paths), curvature (how the world deviates from flatness), and how local measurements control global behavior.
Boundary signal (how the tool fails). *Limit/regularity boundary:* curvature can concentrate, metrics can degenerate, and smooth descriptions can fail.
Tool-switch (what you do next). Move to coordinate-free invariants, or regularize the metric; use local charts (slices) carefully.

3.12 (12) Real / Complex Analysis

(12) Real / Complex Analysis

Scene (what you see). You stop moving rigid objects and start watching *a value field* painted on the sphere: temperature, height, potential. You zoom in closer and closer to a point to see if local change settles into a stable pattern. If it does, the field has a derivative-like behavior.
Name (the math that formalizes it). Analysis is the mathematics of zooming, limits, and controlled approximation.
Allowed moves (what counts as a "legal" interaction). Legal moves are "zooming in" and taking stable limiting behavior; comparing nearby values; smoothing or regularizing to make zoom stable.
Typical tasks (what you practice). Prove continuity and convergence; estimate error; decide when a local linear picture exists; track how local behavior accumulates into global statements.
Boundary signal (how the tool fails). *Limit/regularity boundary:* zoom does not stabilize; singularities appear; convergence fails.
Tool-switch (what you do next). Strengthen estimates, switch convergence mode, or regularize (change representation to make zoom stable).

3.13 (13) Functional Analysis

(13) Functional Analysis

Scene (what you see). Now your "box" is no longer a physical box. It is a *space of signals/functions*. A single point in this space is a whole waveform. Moving one step in this space changes the entire function. You notice that finite-dimensional intuition starts lying to you.
Name (the math that formalizes it). Functional analysis is geometry and linear motion in spaces of functions.
Allowed moves (what counts as a "legal" interaction). Legal moves are operator-actions on functions (transform the entire waveform at once), plus the choice of how to measure distance between functions (a norm).
Typical tasks (what you practice). Understand stability of operators; detect compactness vs non-compactness; describe spectral behavior (how an operator behaves along special directions).
Boundary signal (how the tool fails). *Limit/regularity boundary:* compactness fails; naive "finite box" assumptions break.
Tool-switch (what you do next). Use weaker viewpoints (weak topology), restrict to compact operators, or approximate by finite-dimensional slices on purpose.

3.14 (14) Measure Theory / Integration

(14) Measure Theory / Integration

Scene (what you see). You paint "weight" over regions of the sphere: some regions count more than others. But you also impose a strict box-contract on what regions you are allowed to talk about. Integration is what you get when you add up infinitely many tiny weighted pieces in a controlled way.
Name (the math that formalizes it). Measure theory formalizes "what is a legal region" and "how much it weighs".
Allowed moves (what counts as a "legal" interaction). Legal moves are: build legal regions by closure rules; push weight forward under maps; approximate complicated fields by simple step-like fields.
Typical tasks (what you practice). Decide what is measurable; compute integrals as controlled limits; switch viewpoint by product spaces (joint vs marginal).
Boundary signal (how the tool fails). *Representation boundary:* some subsets cannot be assigned consistent weight under the contract.
Tool-switch (what you do next). Adjust the contract (choose the right sigma-structure), or change representation (simple approximations).

3.15 (15) Probability Theory

(15) Probability Theory

Scene (what you see). You shake the box inside the sphere. You do not choose the next position; the world samples it. Over time, you stop caring about single outcomes and care about the *cloud of positions* (a distribution). You watch how the cloud moves under repeated random steps.
Name (the math that formalizes it). Probability theory is the mathematics of random motion of clouds.
Allowed moves (what counts as a "legal" interaction). Legal moves are stochastic updates (random transitions); combining information by conditioning; projecting a joint cloud to a marginal cloud.
Typical tasks (what you practice). Predict long-run behavior; compare different notions of convergence; compute expectations as weighted averages.
Boundary signal (how the tool fails). *Computational boundary (sometimes):* exact inference can be intractable.
Tool-switch (what you do next). Approximate (Monte Carlo), restrict to simpler model classes (a slice), or summarize by sufficient statistics.

3.16 (16) Stochastic Processes (SDE intuition)

(16) Stochastic Processes

Scene (what you see). You keep shaking continuously. The box does not move smoothly like a sliding block; it jitters at every scale. When you zoom in, the path keeps looking rough. The motion is continuous in time but refuses to have a stable "velocity".
Name (the math that formalizes it). Stochastic processes study random paths and their laws.
Allowed moves (what counts as a "legal" interaction). Legal moves are stochastic evolutions over time; comparing laws of paths; changing filtration (what you are allowed to know at each time).
Typical tasks (what you practice). Describe path regularity, hitting times, long-run distributions, and noise-driven dynamics.
Boundary signal (how the tool fails). *Limit/regularity boundary:* classical smooth calculus breaks on rough paths.
Tool-switch (what you do next). Use the correct stochastic calculus; summarize by finite-dimensional distributions and martingale tools.

3.17 (17) ODE / Dynamical Systems

(17) ODE / Dynamical Systems

Scene (what you see). You stop shaking and instead place a wind field inside the sphere. At every point, the wind tells the box "move this way now". Release a point and it flows along the wind, tracing a trajectory. Some regions pull trajectories in; others repel them.
Name (the math that formalizes it). Dynamical systems formalize flow under deterministic local rules.
Allowed moves (what counts as a "legal" interaction). Legal moves are time-evolution under a rule field; comparison of trajectories; linearize locally (replace a small neighborhood by a straightened box motion).
Typical tasks (what you practice). Find equilibria, cycles, attractors; analyze stability; understand phase portraits.
Boundary signal (how the tool fails). *Limit/regularity boundary:* solutions can cease to exist globally; sensitivity explodes.
Tool-switch (what you do next). Use invariants (energy-like quantities), Lyapunov reasoning, or change coordinates to simplify the flow.

3.18 (18) PDE / Applied Analysis

(18) PDE / Applied Analysis

Scene (what you see). Now the wind is not pushing a single point; it evolves an entire painted field over the sphere at once. The picture can smooth out (like heat) or form sharp fronts (like shocks). Slicing into modes reveals hidden simplicity.
Name (the math that formalizes it). PDE is the mathematics of field evolution.
Allowed moves (what counts as a "legal" interaction). Legal moves are evolution operators on fields; slicing by frequency or symmetry; regularization and weak formulations when smoothness breaks.
Typical tasks (what you practice). Prove existence/uniqueness (when possible), track regularity, understand wave propagation, diffusion, and shock formation.
Boundary signal (how the tool fails). *Limit/regularity boundary:* singularities, shocks, and blow-up.
Tool-switch (what you do next). Switch to weak solutions, add entropy/energy constraints, or use spectral slices (Fourier viewpoint).

3.19 (19) Numerical Analysis

(19) Numerical Analysis

Scene (what you see). You cannot move continuously. You can only take steps on a grid inside the sphere. You approximate smooth motion by a sequence of box-steps. Sometimes the stepper behaves; sometimes it spirals out even when the true motion is stable.
Name (the math that formalizes it). Numerical analysis formalizes the box-approximation of sphere-dynamics.
Allowed moves (what counts as a "legal" interaction). Legal moves are discretizations: replace flow by step rules; replace fields by grids; replace integrals by finite sums.
Typical tasks (what you practice). Decide stability, accuracy, convergence; control error propagation; design robust discretizations.
Boundary signal (how the tool fails). *Representation boundary:* the discretization is a wrong camera—it amplifies error.
Tool-switch (what you do next). Change step size, change scheme, analyze stability regions, or adaptively refine (local zoom).

3.20 (20) Optimization / Variational Methods

(20) Optimization / Variational Methods

Scene (what you see). Inside the sphere there is a height landscape. The box is a hiker sliding downhill. Sometimes the landscape is a simple bowl (one global bottom). Sometimes it is a field of traps (many local bottoms). Constraints act like walls that the hiker cannot cross.
Name (the math that formalizes it). Optimization formalizes motion toward "best" under constraints.
Allowed moves (what counts as a "legal" interaction). Legal moves are descent steps, projected steps (when walls exist), and reparameterizations that reshape the landscape.
Typical tasks (what you practice). Find minima/maxima; prove convergence for nice landscapes; design algorithms for rough landscapes.
Boundary signal (how the tool fails). *Representation boundary:* nonconvexity makes the "downhill" story ambiguous globally.
Tool-switch (what you do next). Change representation, relax constraints, accept local solutions, or use stochastic motion to escape traps.

3.21 (21) Number Theory

(21) Number Theory

Scene (what you see). You replace the smooth sphere-world by a lattice floor. The box can only sit on integer grid points. Sliding by one unit is meaningful; sliding by half a unit is illegal. Patterns repeat in modular cycles, like a clock.
Name (the math that formalizes it). Number theory is the mathematics of lattice motion and modular cycles.
Allowed moves (what counts as a "legal" interaction). Legal moves are integer steps, modular reductions, gcd-like simplifications, and structure-preserving transformations of lattices.
Typical tasks (what you practice). Solve congruences; classify residues; understand multiplicative structure; lift discrete problems into smoother worlds when helpful (analytic tools).
Boundary signal (how the tool fails). *Representation boundary:* continuous intuition fails unless you lift the problem carefully.
Tool-switch (what you do next). Return to invariants (gcd, residue classes) or deliberately lift to a higher viewpoint (embedding into smooth methods).

3.22 (22) Algebraic Geometry

(22) Algebraic Geometry

Scene (what you see). You carve shapes by *polynomial walls.* The box is allowed to move only where all walls agree. The resulting shape may look smooth in some areas and kinked in others. Slicing by a plane reveals simpler cross-sections that still carry essential structure.
Name (the math that formalizes it). Algebraic geometry formalizes shapes defined by polynomial constraints and maps between them.
Allowed moves (what counts as a "legal" interaction). Legal moves are structure-preserving maps between polynomial-cut shapes; taking slices; completing the picture by adding "points at infinity" when needed.
Typical tasks (what you practice). Study dimension, singular points, intersections, and how maps transform solution sets.
Boundary signal (how the tool fails). *Representation boundary:* singularities are places where naive coordinates stop describing the geometry.
Tool-switch (what you do next). Change coordinates, take generic slices, pass to projective completion, or replace point-sets by richer structure data.

3.23 (23) Category Theory (Meta-language)

(23) Category Theory

Scene (what you see). You stop asking about one box and start asking about *translations between boxes*. One branch has its own objects and legal moves. Another branch has a different contract. You want a machine that converts one world into another while preserving composition of moves.
Name (the math that formalizes it). Category theory formalizes "maps between math worlds".
Allowed moves (what counts as a "legal" interaction). Legal moves are: package objects+moves into a category; translate by functors; compare translations by natural transformations.
Typical tasks (what you practice). Build abstractions that reveal common patterns across branches; design systematic tool-switching.
Boundary signal (how the tool fails). *Representation boundary:* a category may be too weak/too strong to capture the structure you want.
Tool-switch (what you do next). Enrich the language (add structure), restrict to a subcategory (slice), or pick a different abstraction level.

3.24 (24) Computability Theory

(24) Computability Theory

Scene (what you see). You build a robot that moves the box by following a program. You ask: can the robot always decide whether it will ever stop moving? Some behaviors resist all possible rule-programs.
Name (the math that formalizes it). Computability theory formalizes what can be decided by algorithms at all.
Allowed moves (what counts as a "legal" interaction). Legal moves are reductions: translate one decision problem into another. You prove impossibility by showing "if you could solve this, you could solve that".
Typical tasks (what you practice). Classify problems as decidable, semi-decidable, or undecidable; design reductions.
Boundary signal (how the tool fails). *Computational boundary:* undecidability (no algorithm exists for all cases).
Tool-switch (what you do next). Change the goal: accept semi-decision, restrict the input class (slice), or move to approximate/heuristic procedures.

3.25 (25) Complexity Theory

(25) Complexity Theory

Scene (what you see). An algorithm exists, but it is too slow. The box can be moved legally, but the number of required micro-moves explodes. So the boundary is not "possible vs impossible" but "possible vs feasible".
Name (the math that formalizes it). Complexity theory formalizes resource limits of computation.
Allowed moves (what counts as a "legal" interaction). Legal moves are efficient reductions and resource accounting. You compare problems by whether one can be solved using another efficiently.
Typical tasks (what you practice). Classify problems by time/space; identify hardness; design approximations.
Boundary signal (how the tool fails). *Computational boundary (pragmatic):* intractability under accepted resource models.
Tool-switch (what you do next). Parameterize (slice the input by a controlling parameter), approximate, randomize, or restrict to special subclasses that are tractable.

4 Part III — Fractal Extension (Multi-scale zoom that preserves the same logic)

Definition 1 (Fractal zoom rule (learning view))**.** When you zoom into a corner/feature, you find another box-in-sphere again. This encodes multi-scale learning:

$$\text{discipline} \to \text{field} \to \text{topic} \to \text{concept} \to \text{exercise type}.$$

At each scale, you still ask: what is the scene, what are legal moves, what is invariant, what boundary is near, and what tool-switch is available.

Practical truncation (avoid explosion). Stop at a fixed depth (usually 3 or 4 levels). The atlas is for navigation, not for enumerating every possible node.

5 Part IV — A Short "How to Study" Protocol (using interaction-first mapping)

Protocol (use this on any chapter).

1. Build the scene: "What is my box? What is my sphere? What do I move?"
2. List legal moves (translation/rotation/squeeze/shear/slice/project/twist/-cut/glue/shake/zoom).
3. Name the invariants you are tracking (what does not change under those moves).
4. Identify the boundary signal (proof, computation, limits, representation).
5. Only then name the branch and collect theorems/techniques as the formal toolset.

Remark 1. If you force the branch name first, you will memorize labels. If you force the scene first, you will recognize tools.

6 Conclusion

Closing. This interaction-first atlas turns the Box–Sphere picture into an actual interface: you see an interaction (slide/rotate/squeeze/shear/slice/project/twist/cut/glue/shake/zoom), and you immediately know which mathematical language owns that interaction.